

UK Power Market Analytics

How can residual demand, renewable generation, and interconnector conditions be used to reason about wholesale electricity price risk?

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Portfolio focus: Data & Analytics • Python • Machine Learning • Business Interpretation

Executive Summary

This project develops a transparent, reproducible workflow for analysing simulated UK electricity-market conditions and estimating wholesale electricity prices. The analysis uses 8,760 hourly observations covering 2025 and examines demand, wind and solar generation, renewable share, residual demand, interconnector imports, and a synthetic fuel-cost proxy.

A time-based 80/20 train-test split was used to compare Linear Regression with Random Forest. On the held-out test period, Linear Regression achieved MAE of £6.68/MWh, RMSE of £9.85/MWh and R^2 of 0.475. Random Forest achieved MAE of £6.93/MWh, RMSE of £10.16/MWh and R^2 of 0.442. The results show meaningful predictive signal, but also indicate that the simulated market is not captured perfectly by the tested feature set.

Scenario analysis illustrates the business interpretation: relative to a representative baseline of £63.43/MWh, a stress scenario produced £76.69/MWh (+20.9%), while a flexibility scenario produced £52.73/MWh (−16.9%). These scenario outputs are model estimates within the simulated environment, not forecasts of the real UK electricity market.

1. Business Question

How can residual demand, renewable generation, and interconnector conditions be used to reason about wholesale electricity price risk?

The business relevance is straightforward: tighter supply-demand conditions can increase exposure to wholesale price volatility, while additional renewable output and cross-border import capacity can change the amount of conventional generation required. The project therefore focuses on relationships that a commercial or strategy team could use to structure market-risk discussions.

2. Analytical Approach

Dataset: synthetic hourly observations for 2025 (8,760 rows). Inputs: demand, wind, solar, renewable share, residual demand, interconnector imports, gas-price proxy, and time features. Target: wholesale electricity price (£/MWh).

Feature engineering: residual demand was calculated as demand minus wind and solar generation. Cyclical encodings were added for hour, day of week, and month to represent recurring temporal patterns.

Validation: the first 80% of observations were used for training and the final 20% for testing, preserving chronological order. This avoids randomly mixing future observations into the training sample.

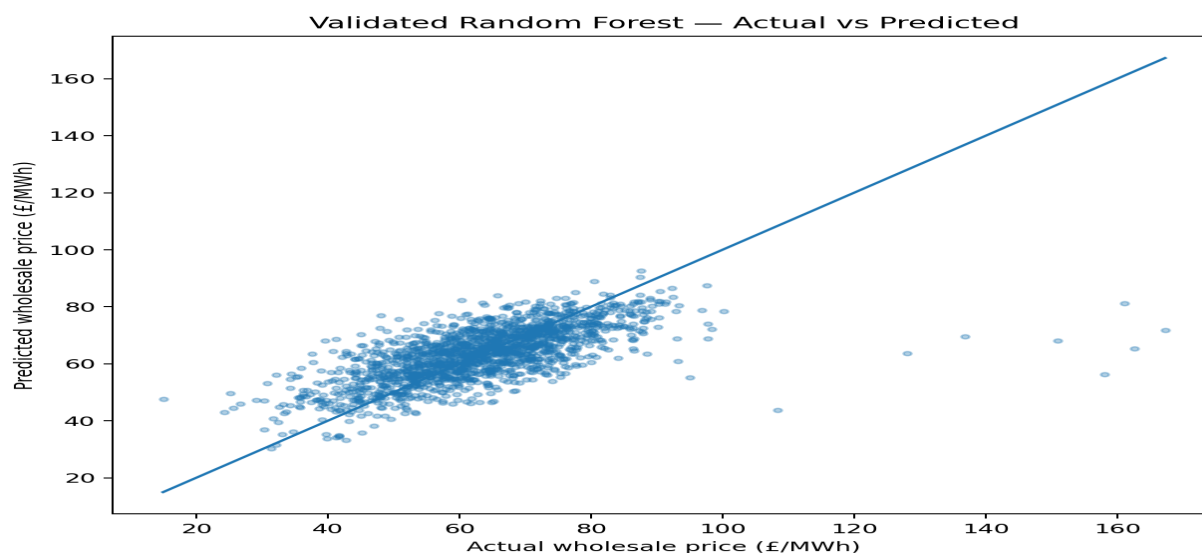
Models: Linear Regression provides an interpretable baseline; Random Forest provides a nonlinear comparison and feature-importance output.

3. Model Results

Model	MAE (£/MWh)	RMSE (£/MWh)	R ²
Linear Regression	6.68	9.85	0.475
Random Forest	6.93	10.16	0.442

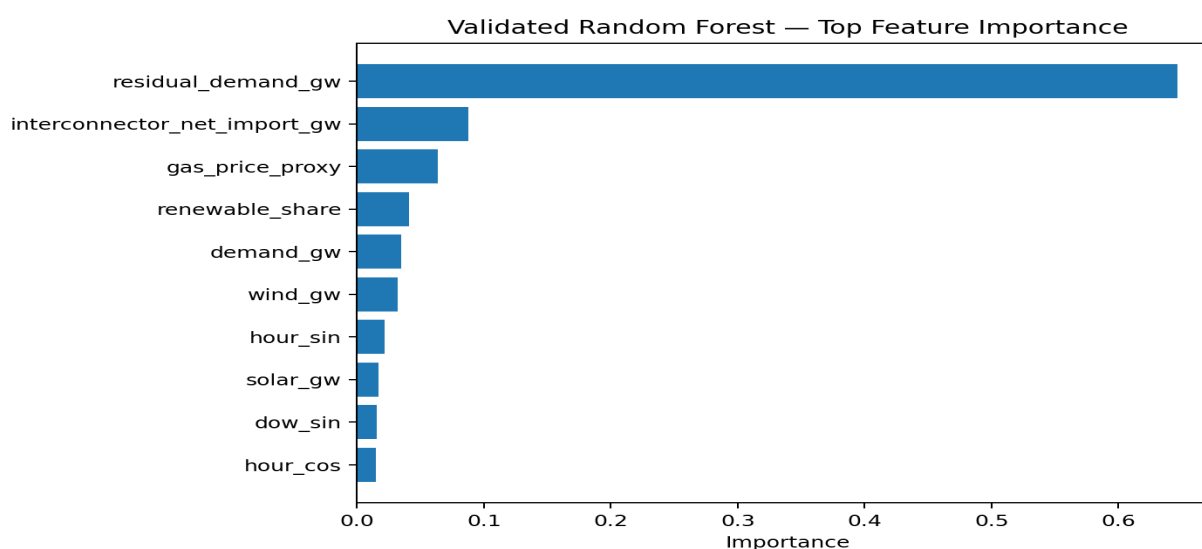
Interpretation: the Linear Regression baseline slightly outperformed Random Forest on the held-out period. This is itself an analytical finding: the additional nonlinear model did not improve predictive accuracy under this feature set and simulated data-generating process. The project therefore avoids claiming that Random Forest is inherently superior.

Actual vs Predicted



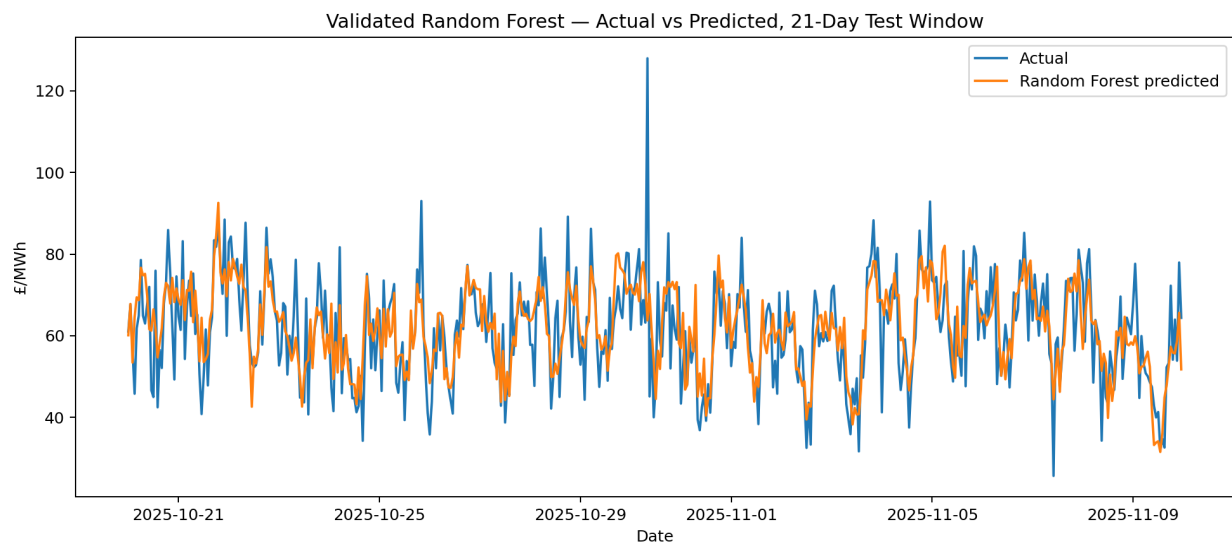
The scatter plot compares held-out actual prices with Random Forest predictions. Points closer to the diagonal indicate smaller prediction errors.

Feature Importance



Residual demand is the dominant Random Forest feature, followed by interconnector imports and the gas-price proxy. Importance indicates model reliance, not causality.

21-Day Test Window



The time-series view shows how predicted prices track the direction and level of actual prices across an initial 21-day window of the test period.

4. Scenario Analysis

Scenarios were applied to a representative test-period operating point. The model was then asked to estimate the price under changed market conditions. This is a sensitivity exercise, not a forecast or causal simulation.

Scenario	Modelled Price	vs Baseline
Baseline	£63.43	—
Stress - higher demand / lower renewables	£76.69	+20.9%
Flexibility - lower demand / higher renewables	£52.73	-16.9%

The stress case combines 10% higher demand, 20% lower wind and solar generation, and 1 GW lower interconnector imports. The flexibility case combines 5% lower demand, 20% higher wind and solar generation, and 1 GW higher imports.

In the simulated environment, the stress configuration raises the modelled price by 20.9%, while the flexibility configuration reduces it by 16.9%. The result demonstrates how the model can be used as a structured sensitivity tool for commercial discussion.

5. Key Findings

- Residual demand is the strongest feature in the Random Forest model, consistent with the project's focus on the remaining system load after renewable generation.
- Interconnector imports and the gas-price proxy are also among the most influential model inputs.
- Linear Regression slightly outperformed Random Forest on the held-out test period, so the project does not claim that the more complex model is automatically better.
- Scenario analysis provides a practical way to translate model inputs into business-facing sensitivity estimates.
- The analysis is useful as a portfolio demonstration of data preparation, modelling, validation, visualisation and commercial interpretation.

6. Limitations & Next Steps

Simulated data: the dataset is synthetic and is not a representation of live UK market observations. The model metrics therefore should not be interpreted as real-market forecasting accuracy.

Model scope: the feature set does not include weather forecasts, actual fuel curves, transmission constraints, outages, market bidding behaviour, policy events, or other drivers that can influence real wholesale prices.

Next step: replace the synthetic dataset with historical market and system data, perform rolling or walk-forward validation, test additional algorithms, and conduct sensitivity and explainability analysis.

7. Reproducibility

The project workbook contains the simulated data sample, data dictionary, model results, feature importance, test predictions and scenario analysis. The outputs were generated with a fixed random seed (42) so the synthetic dataset can be reproduced consistently.

Evidence boundary: all numerical model and scenario results in this document are outputs from this project run. They are not copied from the earlier draft case study.